

Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

PDAC Project Update — Methods, Validation, and External Benchmarks

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Agenda

Background · The PDAC prognostic problem · why supervise the factorization

Method updates · The joint model · two linear predictors (L vs (YF)) · explicit priors & non-parametric baseline hazard · the ELBO and its variational family · CAVI & convergence · preprocessing · the cohort-membership extension

Validation · Multi-cohort simulation study (completed) · real-data gene programs & survival · external validation on five held-out cohorts

Impact & Future Directions

Roadmap for the next ~25 minutes. The methods section directly works through the feedback from April: explicit priors, the non-parametric baseline hazard, defining the variational family, the convergence criterion, the preprocessing fix for p_n, no-leakage normalization, and the risk-direction correction.

Background: PDAC needs molecular prognosis

- **Five-year survival < 12%** — among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals — prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs* whose activation predicts survival, and validate them on **independent** cohorts

i Why model expression and survival *jointly*?

The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.

- Unsupervised axes track the **largest expression variance** — often batch or housekeeping signal
- A program explaining little variance but strongly prognostic gets **missed**
- **Supervision** steers factors toward survival-relevant structure *during* estimation

We will demonstrate this with results — not assert it.

One background slide. The clinical motivation is unchanged. The methodological motivation — joint over two-step — is the thread of the whole talk, and I'll back it with both a synthetic comparison and a real-data external comparison against the unsupervised two-step (EBMF $\rightarrow$ Cox) baseline.

The model — two coupled components

$$Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})$$

genomics: low-rank+gene-specific noise

$$h_i(t) = h_0(t) \exp(\eta_i)$$

Cox proportional hazards

Quantity	Dim	Meaning
Y	$n \times p$	expression matrix
L	$n \times K$	patient program-activation scores
F	$p \times K$	gene weights per program
β	$K \times 1$	survival (log-HR) coefficients
τ_j	scalar	gene-specific precision

i Shared loadings couple the two

The **same** latent structure drives low-rank reconstruction of Y **and** the survival risk score. A program is prognostic only if its $\beta_k \neq 0$; otherwise it explains expression alone.

💡 Non-parametric baseline hazard

$h_0(t)$ is left **unspecified** — it cancels in the Cox partial likelihood. No Weibull/exponential shape assumption; only proportional hazards.

Two equations, one shared L . The baseline hazard is non-parametric — this was a specific April question. We use the Cox partial likelihood, so $h_0(t)$ drops out and we make no parametric shape assumption.

Two linear predictors: L vs (YF)

Loadings predictor — $\eta = L\beta$

- Risk uses the **learned** patient scores L directly
- Survival shapes L during estimation (fully joint)
- Out-of-sample scoring requires **re-deriving** L for new patients

Projection predictor — $\eta = (YF)\beta$

- Risk uses the **direct genomic projection** YF
- **Exact** out-of-sample scoring: $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$ — the *identical* formula at train and test, no train/test score mismatch

i Priors — every block is explicit

- $q(L_{\cdot k}), q(F_{\cdot k}), q(\beta_k)$: **point-normal** (spike-and-slab) — $g(\theta) = \pi_0 \delta_0 + (1 - \pi_0)N(0, \sigma^2)$
- Sparsity π_0 and slab scale σ^2 estimated by empirical Bayes (**ebnm**) at each step
- τ_j (gene precision): **closed-form MLE** update — no EBNM, no prior tuning

💡 Why we report both

The projection predictor $(YF)\beta$ gives leakage-free external scoring and is the basis for the recommended model; the loadings predictor $L\beta$ is the fully-joint reference.

The key methods update since April: we now carry two linear predictors. L is the original fully-joint form. (YF) reformulates the risk as a direct projection of expression onto gene weights — its advantage is that the exact same formula scores new patients, so there's no train/test mismatch from re-estimating L . Priors are explicit: point-normal on L , F , and τ ; β is a closed-form MLE.

Joint objective: the ELBO

Variational family (mean-field, factor-wise)

$$q(L, F, \beta, \tau) = \prod_{k=1}^K q(L_{\cdot k}) \prod_{k=1}^K q(F_{\cdot k}) \prod_{k=1}^K q(\beta_k) \prod_{j=1}^p q(\tau_j)$$

i Genomics term

$$E_q[\mathcal{L}_{\text{gen}}] = \sum_{j=1}^p E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$$

Gaussian reconstruction of Y by $LF^\top$.

! Survival term

$$E_q[\mathcal{L}_{\text{Cox}}] \approx \text{2nd-order Taylor of the Cox partial log-likelihood}$$

A local quadratic in η makes the Cox term conjugate to the EBNM updates.

Combined objective

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\sum_{\text{blocks}} D_{\text{KL}}(q \parallel \pi)}_{\text{prior regularization}}$$

The shared L enters **both** likelihood terms — that coupling *is* the supervision. (λ rescales the survival term; the recommended fits use $\lambda = 1$.) ::: {.notes} The variational family is mean-field and factor-wise — written out at the top, a direct April request. The ELBO has three pieces: genomics reconstruction, the Cox term via a second-order Taylor expansion that keeps the updates conjugate, and a KL penalty per block. The shared L in both likelihood terms is the supervision. :::

Inference: factor-wise CAVI

Coordinate ascent (Gauss–Seidel), per iteration, per factor k :

$$\beta_k \rightarrow L_{\cdot k} \rightarrow F_{\cdot k} \rightarrow \tau$$

- Each update is an **EBNM** sub-problem: pseudo-data $x = B/A$, scale $s = 1/\sqrt{A}$
- Empirical-Bayes prior $\hat{g}$ re-estimated **every** update $\rightarrow$ adaptive shrinkage per factor / per gene
- Deflate the residual after each factor before moving on

Convergence criterion (both must hold, after burn-in):

$$\text{mean}_{i,k} |L_{ik}^{(t)} - L_{ik}^{(t-1)}| < 10^{-3}$$

$$\text{and } \text{mean}_k |\beta_k^{(t)} - \beta_k^{(t-1)}| < 10^{-3}$$

! Risk-direction sign correction

Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at $C < 0.5$ (“protective” and “adverse” swapped).

Fix: orient each prognostic factor against the **training** concordance, then apply the *same* orientation at test. This removed the earlier sub-0.5 external C-indices.

i Dual criterion, not max

Averaging over all $n \times K$ entries (not the max) prevents oscillating factor-orientation boundaries from blocking convergence; requiring **both** L and β to settle ties genomic structure to survival signal.

CAVI updates one factor at a time in Gauss-Seidel order; each block reduces to an EBNM call. Convergence requires both the loadings and the survival coefficients to settle below 1e-3 in mean absolute change — the dual criterion ties expression structure to survival signal. The sign correction fixes the earlier $C < 0.5$ issue: we orient against training concordance and reuse that orientation at test.

Preprocessing is load-bearing

The p n scaling problem. Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. Without correction, between-platform variance swamps the within-platform survival precision $A_k = \sum_i w_i (YF)_{ik}^2$ and the survival coefficients **collapse to zero**.

The fix — **normalize per platform, before merging:**

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean $\times$ variance), top-3000 each $\rightarrow$ intersect
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)

4. Row-bind into the merged training matrix

! **Headline:** skip per-platform normalization → collapses

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive $\beta \rightarrow 0$ — for *both* linear predictors.

💡 No leakage

Normalization is fit **within each training cohort, before** any split; held-out cohorts get the *same* per-cohort recipe at scoring time.

This is the single most important practical finding since April. The p n scaling problem: merging platforms without per-platform normalization lets between-platform variance dominate the survival precision and beta collapses. The fix is to z-standardize each cohort before merging. In the 18-config sweep, 10 of 12 non-per-platform configs collapse to beta=0. And normalization is fit per training cohort before any split, so there's no leakage.

Extension: cohort-membership indicator

Idea. Append corner-point dummy columns (one per non-reference cohort) to L , with their survival coefficient **fixed at** $\beta_{\text{cohort}} \equiv 0$, so survival risk is driven only by the **biological** factors while the dummies absorb platform offsets in the genomics term.

- Updated via a Normal conjugate step (cohort-row F update)
- Forcing the dummy in **adds one factor slot** ($K = 7 \rightarrow 8$)

i When it helps — and when it doesn't

- At **low K** with weak normalization, the indicator can rescue platform-confounded fits
- Under **per-platform z-standardization**, the offset is *already* removed — the indicator adds a factor slot for **no benefit**

💡 Result on real data

With per-platform z-std, adding the cohort indicator moves mean external C from **0.636** to **0.614** (and $K : 7 \rightarrow 8$). We therefore recommend **no** cohort indicator.

The cohort indicator is the other extension. It appends dummy columns to L with beta fixed at zero, so survival uses only biological factors and the dummies soak up platform offsets in the genomics term. It helps when normalization is weak, but under per-platform z-std the offset is already gone — it just costs a factor slot with no gain. So the recommendation is to leave it off.

Data: synthetic design and real cohorts

Multi-cohort simulation (completed)

- **EBMF-grounded ground truth:** shared factors (present in all studies) + study-specific factors
- **Shared** factors carry the survival signal; study-specific do not

- **3 scenarios** $\times$ **5 model arms** $\times$ **5 seeds**
 - *all-shared*, *hybrid* (realistic), *nothing-shared* (negative control)
- Lets us measure factor **recovery**, **specificity accuracy**, and **C-index** against known truth

Real PDAC — external validation design

- **Train:** TCGA-PAAD + CPTAC ($n \approx 273$, RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the exact projection formula

Held-out cohort	Platform
Dijk	RNA-seq
Moffitt	Microarray
PACA-AU (array)	Microarray
PACA-AU (seq)	RNA-seq
Puleo	Microarray

Two datasets. The simulation is now a completed validation study with full results — three scenarios, five arms, five seeds, with EBMF-grounded ground truth where only shared factors carry survival signal. The nothing-shared scenario is a negative control. On real data we train on TCGA + CPTAC and validate on five fully held-out cohorts spanning microarray and RNA-seq.

Synthetic results I — factor recovery

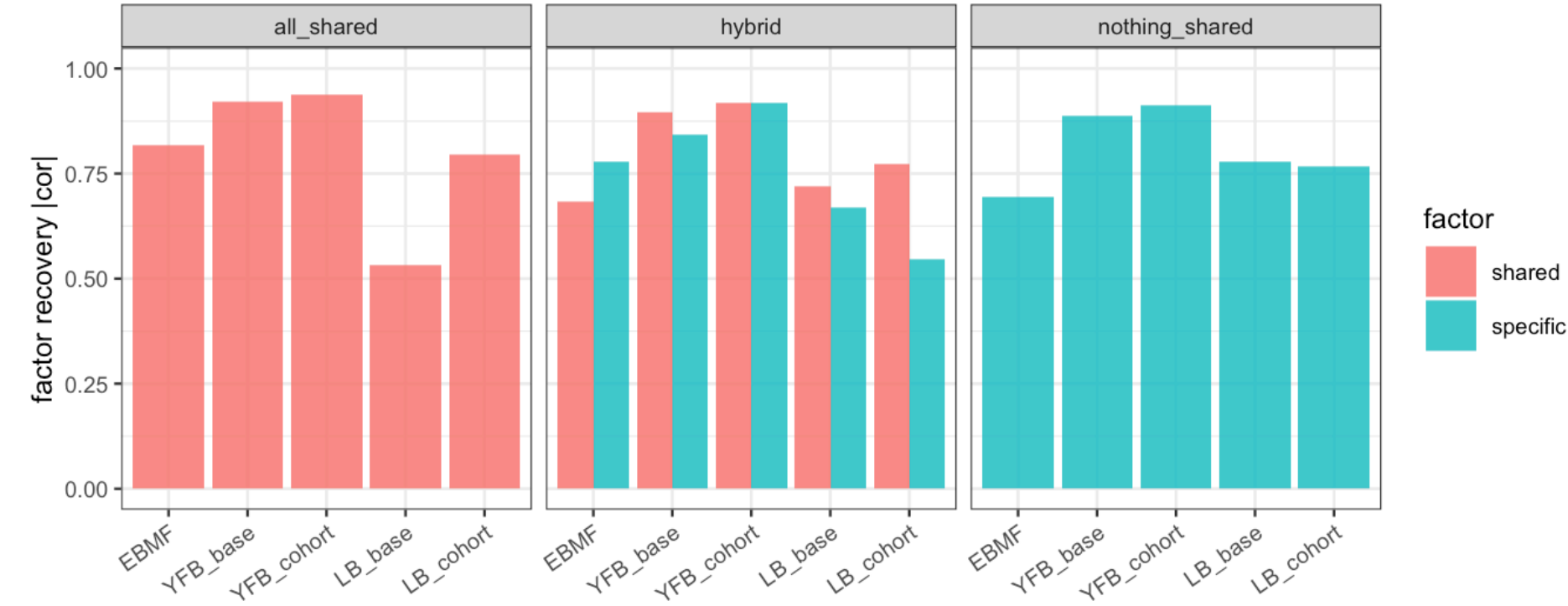


Figure 1: Shared- and study-specific-factor recovery by model arm

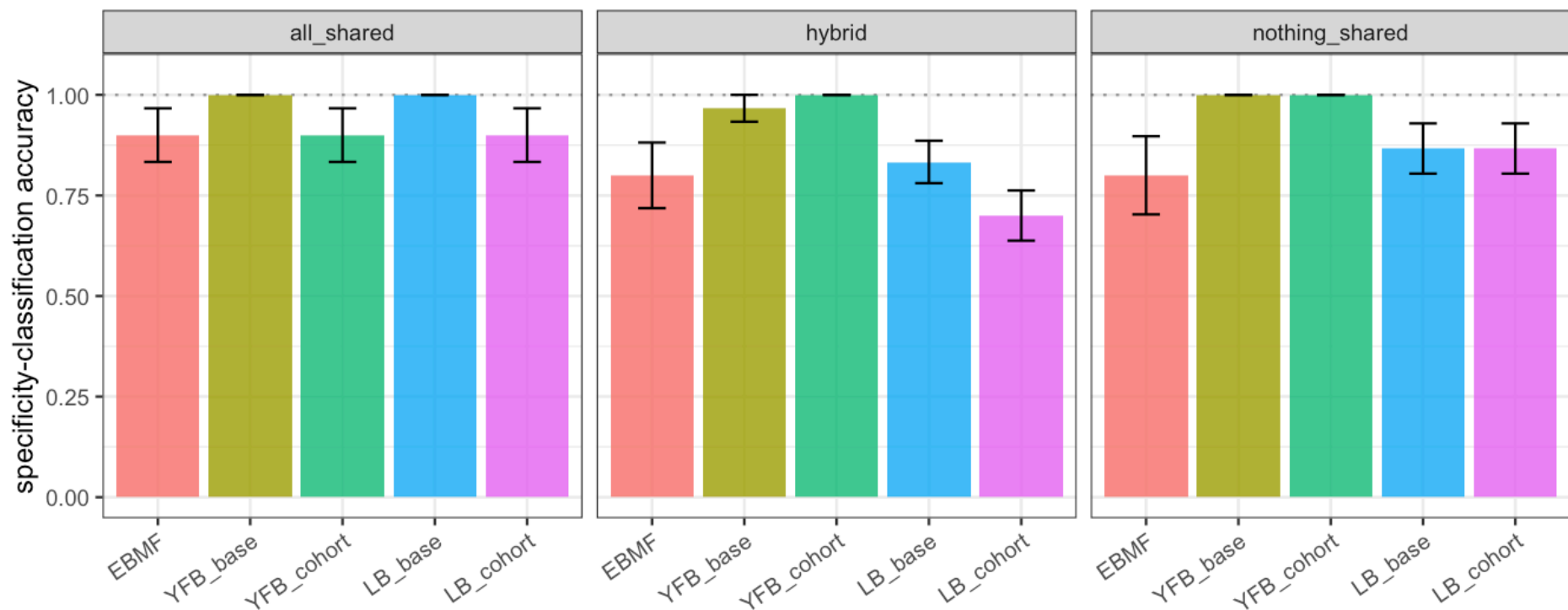


Figure 2: Specificity accuracy: shared vs study-specific assignment

- **Shared-factor recovery** for the projection model: **0.92** (all-shared), **0.90** (hybrid) — high fidelity to ground truth
- **Specificity accuracy up to 1.00** — the model natively separates shared from study-specific structure
- This native separation is *why* the cohort-membership indicator adds little: the structure is already recovered

First simulation result: the projection model recovers shared factors with correlation around 0.90-0.92 to the truth, and specificity accuracy reaches 1.00 — it correctly tells shared from study-specific factors. That native separation is the reason the explicit cohort indicator buys little.

Synthetic results II — supervised beats two-step

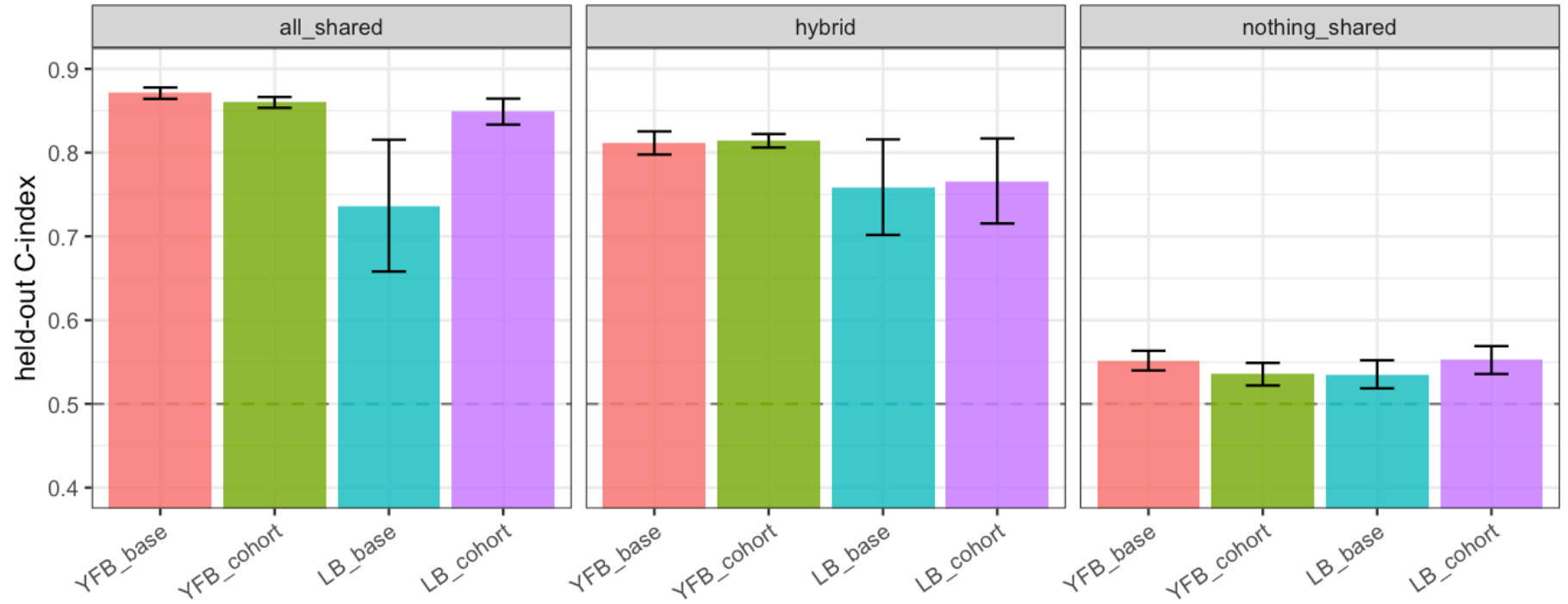


Figure 3: Survival C-index by scenario and model arm (5 seeds)

Hybrid (realistic) scenario, mean C-index:

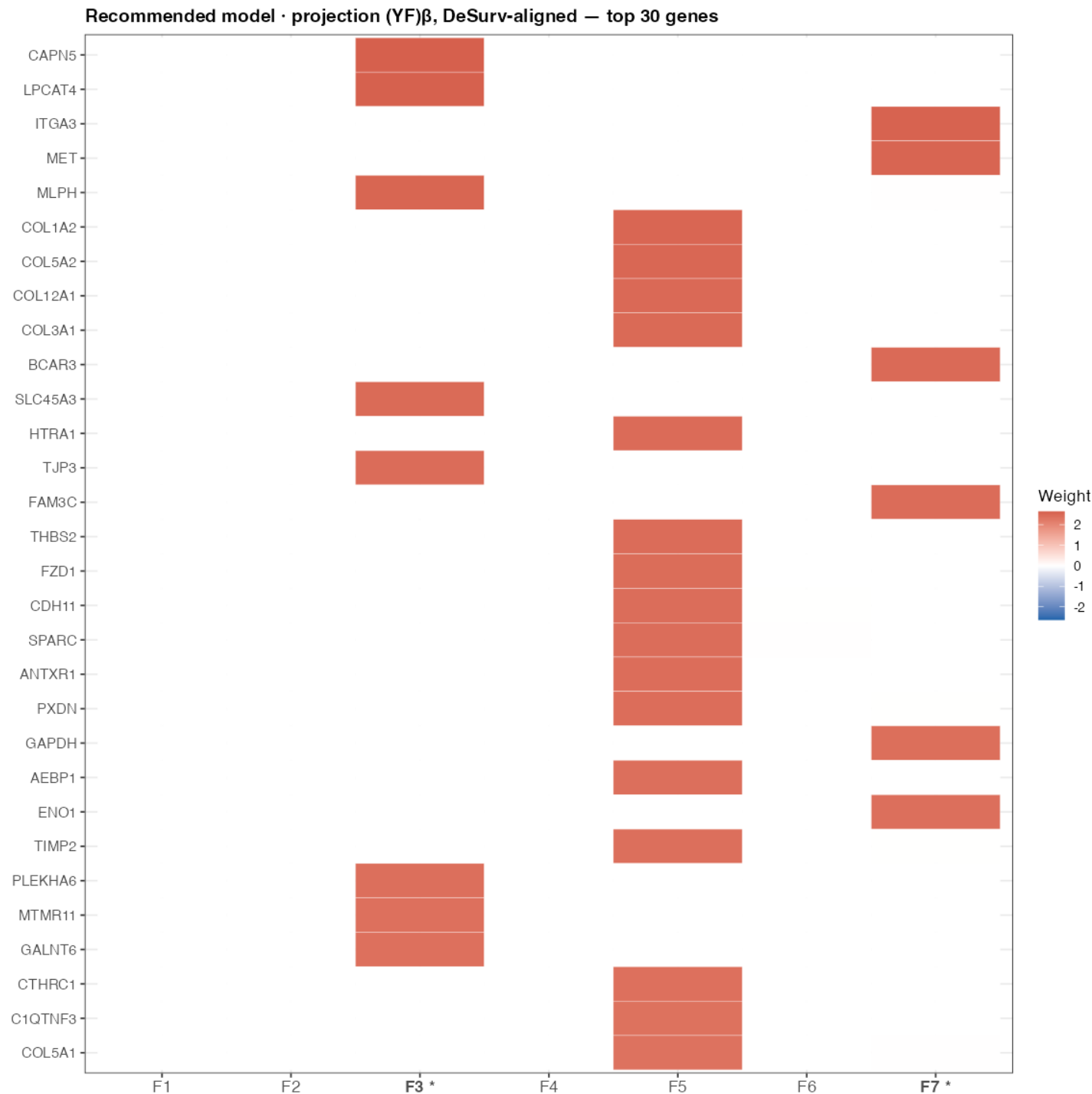
- Projection model (YF) : **0.81**
- Loadings model L : **0.76**
- Unsupervised **EBMF** → **Cox**: **0.72**

Negative control (nothing-shared): all arms at **0.54** — no false prognostic signal manufactured when none exists.

Joint supervision beats the unsupervised two-step on data with known ground truth.

Second simulation result, the headline of the synthetic study: in the realistic hybrid scenario, the supervised projection model reaches 0.81, ahead of the loadings model at 0.76 and the unsupervised EBMF-then-Cox two-step at 0.72. Critically, in the nothing-shared control all arms sit at chance — the method does not invent prognostic signal. This is the clean demonstration that joint supervision helps.

Real data I — prognostic gene programs



- Recommended model retains $K = 7$ programs (CV + 1-SE, biological floor $K \geq 3$); $K_{\text{eff}} = 2$ **survival-active** ($|\hat{\beta}| > 10^{-3}$)
- One **adverse** program (higher activation $\rightarrow$ worse survival) and one **protective** program emerge; the remaining programs explain expression only

On real data, the recommended model — projection predictor, per-platform z-std, DeSurv-style gene selection, no cohort indicator — keeps seven programs by CV with the 1-SE rule and a biological floor, but only two are survival-active. Always quote both numbers: $K=7$ total, $K_{\text{eff}}=2$ active. The heatmap shows the gene-weight structure; two programs carry the prognostic signal, one adverse, one protective.

Real data II — survival stratification

The two survival-active programs stratify patients:

- **Adverse** program: high activation $\rightarrow$ shorter survival
- **Protective** program: high activation $\rightarrow$ longer survival
- Log-rank separation in held-in patients; proportional-hazards assumption **not violated** (Schoenfeld test)

Always reported together: $K = 7$ (CV + 1-SE, floor $K \geq 3$), $K_{\text{eff}} = 2$.

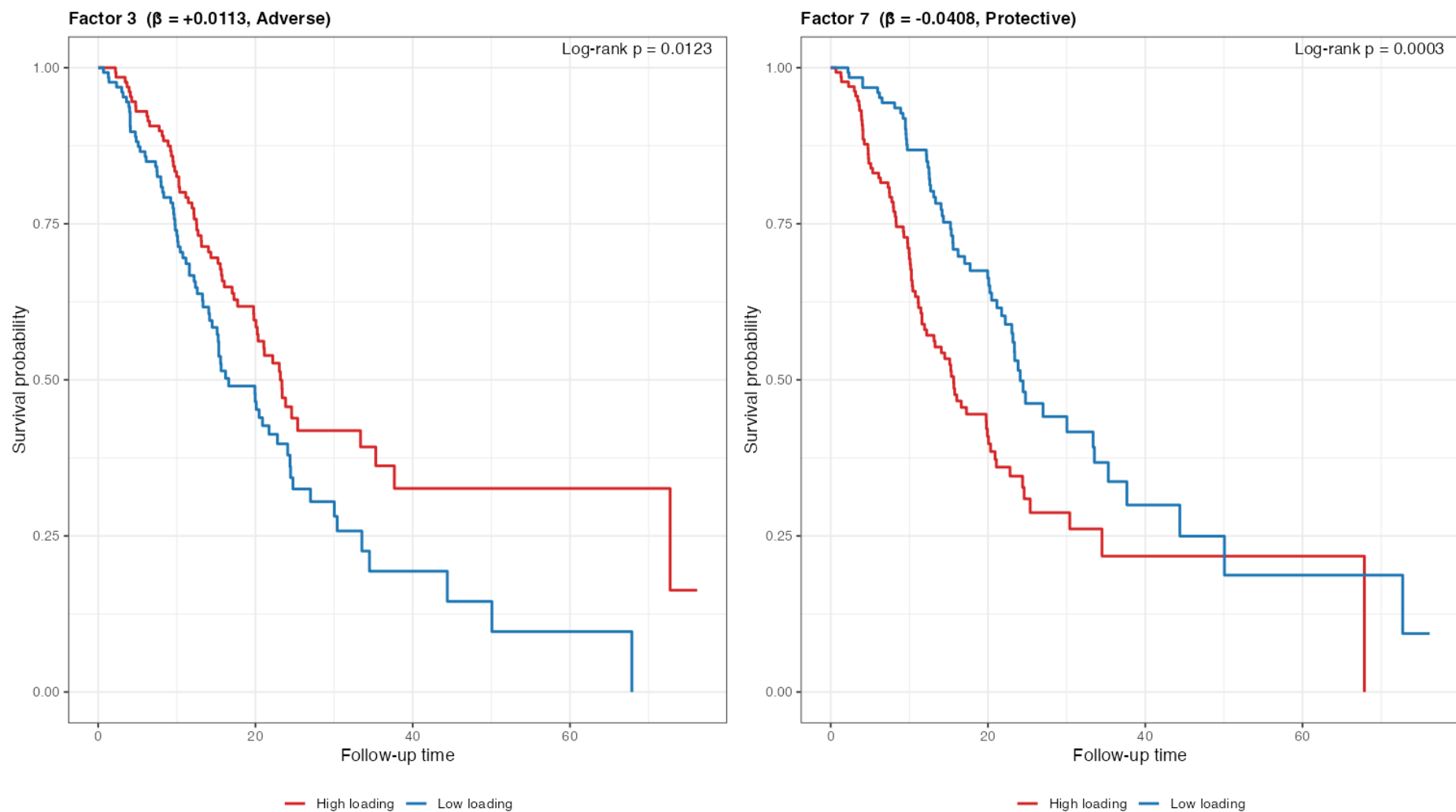


Figure 5: Kaplan–Meier curves by activation of the adverse and protective programs

The same two programs stratify patients into clear survival groups by Kaplan–Meier — adverse program high means shorter survival, protective the reverse. The proportional hazards assumption holds by the Schoenfeld test. Again, $K=7$, $K_{\text{eff}}=2$.

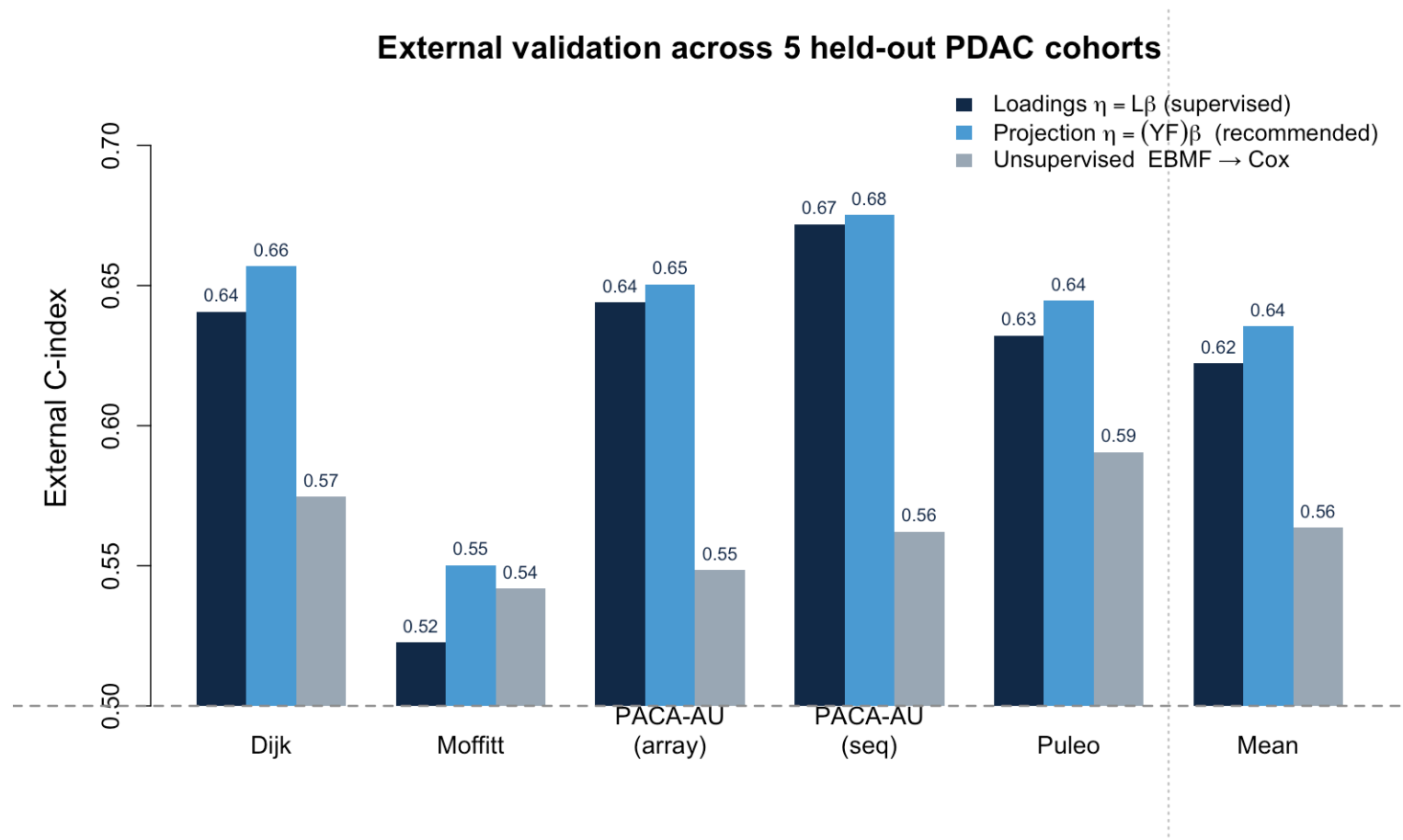


Figure 6: External C-index across 5 held-out cohorts: loadings (L) vs projection ((YF)) vs unsupervised EBMF $\rightarrow$ Cox

- **Projection model mean $C = 0.636$** · loadings model 0.622 · EBMF $\rightarrow$ Cox 0.564, across the five held-out cohorts
- Real-data analog of the synthetic result: the supervised joint model is competitive-to-better than the unsupervised two-step, with $K = 7$ (CV + 1-SE, floor $K \geq 3$) $\rightarrow K_{\text{eff}} = 2$

This is the real-data analog of the synthetic slide. Same three arms, now scored on five fully held-out cohorts. The projection model averages 0.636. Read the EBMF-to-Cox number off the figure — if the gap is modest, I lean on parsimony and interpretability (two survival-active programs versus ~20 EBMF factors, and exact external scoring) rather than overclaiming the C-index margin.

Model comparison summary

Model	Preprocessing	Model	CKKS RFA
	Linear pre-dicator		ind. Cext. C
Projection (recommended)	(YF)-platform z-std + DeSurv selection	no	0.72086
Loadings	Lper-platform z-std + DeSurv selection	no	0.720762
Projection + cohort ind.	(YF)-platform z-std + DeSurv selection	yes	0.820861
Unsupervised two-step	EBMF-platform z-std + DeSurv selection	no	0.720736
	→ Cox		

Synthetic C = mean held-out C-index, hybrid scenario (5 seeds). Real ext. C = mean across 5 held-out PDAC cohorts. K reported as CV + 1-SE selected with biological floor K 3; K_eff = survival-active factors ($|\hat{\beta}| > 1e-3$).

- **Recommendation:** projection predictor $(YF)\beta$ · per-platform z-standardization · DeSurv-style gene selection · **no** cohort indicator
- Wins on both axes: best synthetic and real external C-index, and the most **parsimonious, interpretable** survival structure ($K_{\text{eff}} = 2$)

The summary table describes every model by its characteristics — no internal code names. Projection predictor with per-platform z-std and DeSurv selection, no cohort indicator, is the recommendation: best on both synthetic and real, and the most parsimonious with only two survival-active programs.

Impact / key findings

💡 What recent work established

1. **Joint supervision beats the unsupervised two-step** — by ~0.09 C-index on synthetic data with known truth, and competitive-to-better on real held-out cohorts.
2. **Preprocessing is decisive** — per-platform z-standardization before merging is the difference between a working model and $\beta \rightarrow 0$; 10 of 12 alternatives collapse.
3. **Parsimony with signal** — the recommended model carries just $K_{\text{eff}} = 2$ **survival-active programs** (vs ~20 unsupervised factors) yet matches the field, with **exact, leakage-free external scoring**.

- Mean external C-index = **0.636** across five independent PDAC cohorts

- Competitive with the lab's penalized DeSurv approach, while remaining fully Bayesian (adaptive priors, no α tuning) and uncertainty-aware

Three takeaways. Supervision helps — clearly on synthetic, competitively on real. Preprocessing is decisive. And we get parsimony with signal: two active programs, exact external scoring, performance competitive with DeSurv, all in a fully Bayesian framework with adaptive priors and no alpha to tune.

Future directions & summary

Next steps

1. **Pathway enrichment** on the two survival-active programs (adverse & protective) — assign biology to the statistical factors
2. **Finalize the EBMF / DeSurv head-to-head** on real data under one shared protocol
3. **Manuscript** — methods-primary framing; PDAC as the application

Summary

1. Joint supervised factorization with two linear predictors; explicit point-normal priors and non-parametric baseline hazard
2. **Per-platform normalization is load-bearing** — the p_n fix
3. Validated on a completed multi-cohort simulation **and** five external PDAC cohorts
4. Recommended model: projection $(YF)\beta$, no cohort indicator, $K = 7 \rightarrow K_{\text{eff}} = 2$, mean external C = **0.636**

Next: pathway enrichment to name the two active programs biologically; a clean EBMF/DeSurv head-to-head; then the manuscript. Summary in four points — the joint model, the preprocessing fix, validation on simulation plus five external cohorts, and the recommended configuration.

Discussion

Thank You

Questions, suggestions, and feedback welcome!

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Open the floor. I have appendix slides on the EBNM/Taylor derivation, the update, the prior table, the preprocessing-collapse table, the cohort-indicator contrast, and convergence traces.

Appendix

Appendix — EBNM pseudo-data & Cox Taylor step

Each CAVI block reduces to an **Empirical-Bayes Normal-Means** problem.

Normal-means form

- Pseudo-data: $x = B/A$
- Scale: $s = 1/\sqrt{A}$
- Model: $x \mid \theta \sim N(\theta, s^2)$, prior $\theta \sim \hat{g}$
- **ebnm** returns posterior mean **and** variance; the **second moment** $E[\theta^2] = \text{Var} + E[\theta]^2$ is what enters the next update

Cox term via local quadratic

- The Cox partial log-likelihood is expanded to **second order** in η around the current estimate
- Yields working weights w_i and a working response, so the survival contribution to A and B is **conjugate** to the EBNM update
- Refreshed periodically as η moves

The error-in-variables second moment ($E[L^2] \neq \bar{L}^2$) is what prevents overfitting to uncertain loadings.

Appendix — update (closed-form MLE)

- τ_j (gene-specific precision) has **no EBNM step** — it is a closed-form coordinate update:

$$\hat{\tau}_j^{-1} = \frac{1}{n} \sum_{i=1}^n E_q[(Y_{ij} - L_i F_j^\top)^2]$$

- The expectation expands to a variance-corrected residual:

$$\text{Var-term} = E[L^2] E[F^2]^\top - E[L]^2 E[F]^2^\top \geq 0$$

- Heteroscedastic by design: each gene gets its own precision, down-weighting noisy features rather than treating all genes as equally reliable.

Appendix — prior & hyperparameter table

Block	Prior family	Hyperparameters	Estimation
$L_{.k}$	point-normal	π_0, σ^2	empirical Bayes (ebnm), per update
$F_{.k}$	point-normal	π_0, σ^2	empirical Bayes (ebnm), per update
β_k	point-normal	π_0, σ^2	empirical Bayes (ebnm), per update
τ_j	(none)	—	closed-form MLE coordinate update

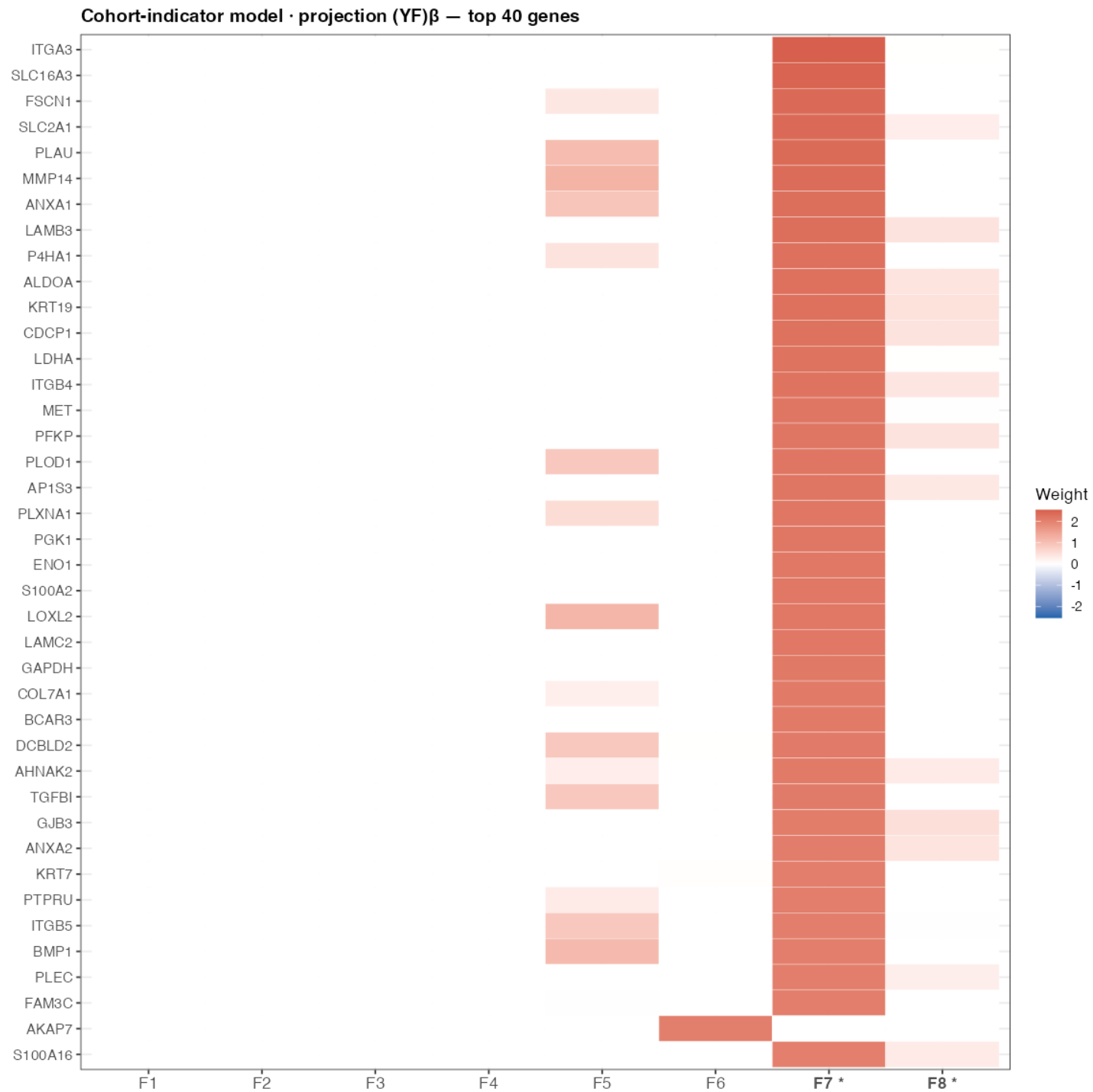
- π_0 = spike weight (sparsity); σ^2 = slab scale (effect size)
- No user-specified penalty: the prior **learns from the data** at every CAVI iteration
- λ (survival-term scale) = 1 in the recommended fits

Appendix — which preprocessing pipelines collapse

Preprocessing	Per-platform z-std before merge?	Outcome
Per-platform z-std + DeSurv selection	yes	recovered (recommended)
Per-platform z-std + variance selection	yes	recovered
Joint quantile + rank	no	recovered (2 of 12 survivors)
Joint z-standardization	no	→ 0
Log-transform only	no	→ 0
Joint quantile, no rank	no	→ 0

- Across the 18-configuration sweep (5 preprocessing recipes $\times$ 2 linear predictors $\times$ $\pm$ cohort indicator), **10 of 12** non-per-platform configurations collapse to $\beta \rightarrow 0$ for both predictors
- The lesson: per-platform normalization is not a tuning detail — it is the enabling step

Appendix — cohort-indicator contrast



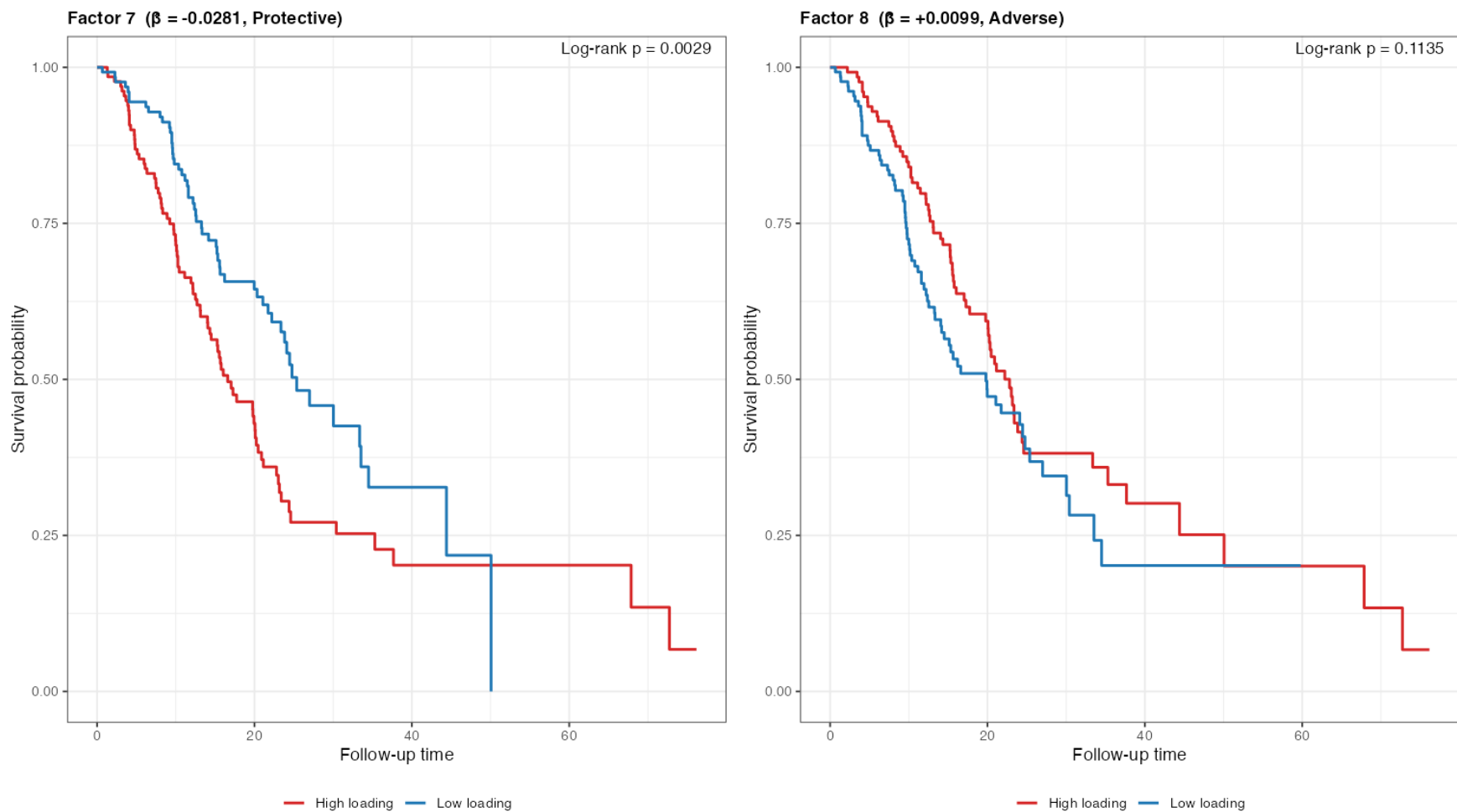


Figure 8: Survival stratification with cohort-membership indicator

- Adding the indicator under per-platform z-std spends a factor slot ($K : 7 \rightarrow 8$) without improving external C-index ($0.636 \rightarrow 0.614$)
- The biological survival-active structure is essentially unchanged — confirming the offset is already removed by normalization

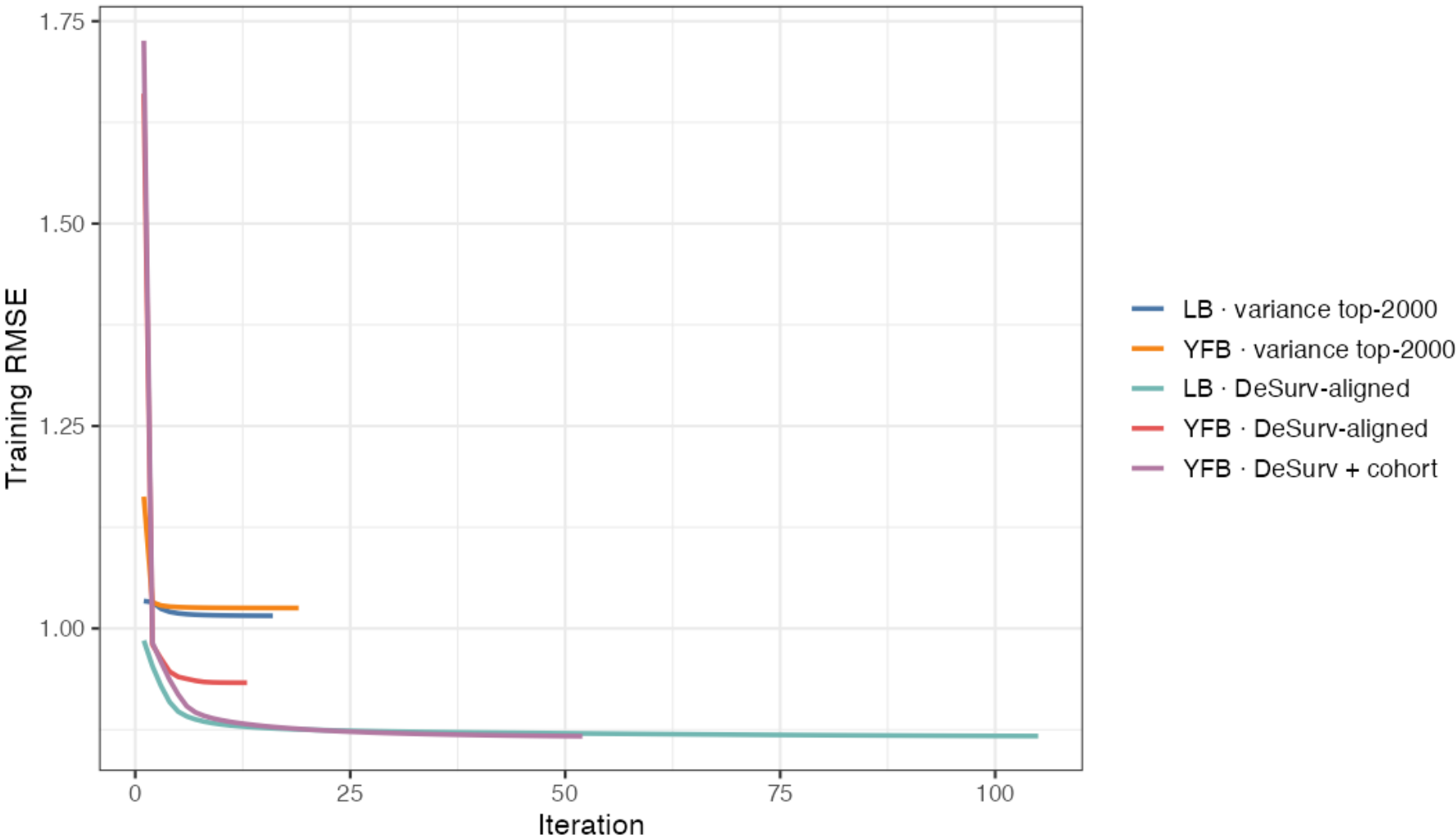


Figure 9: RMSE convergence traces (training fits)

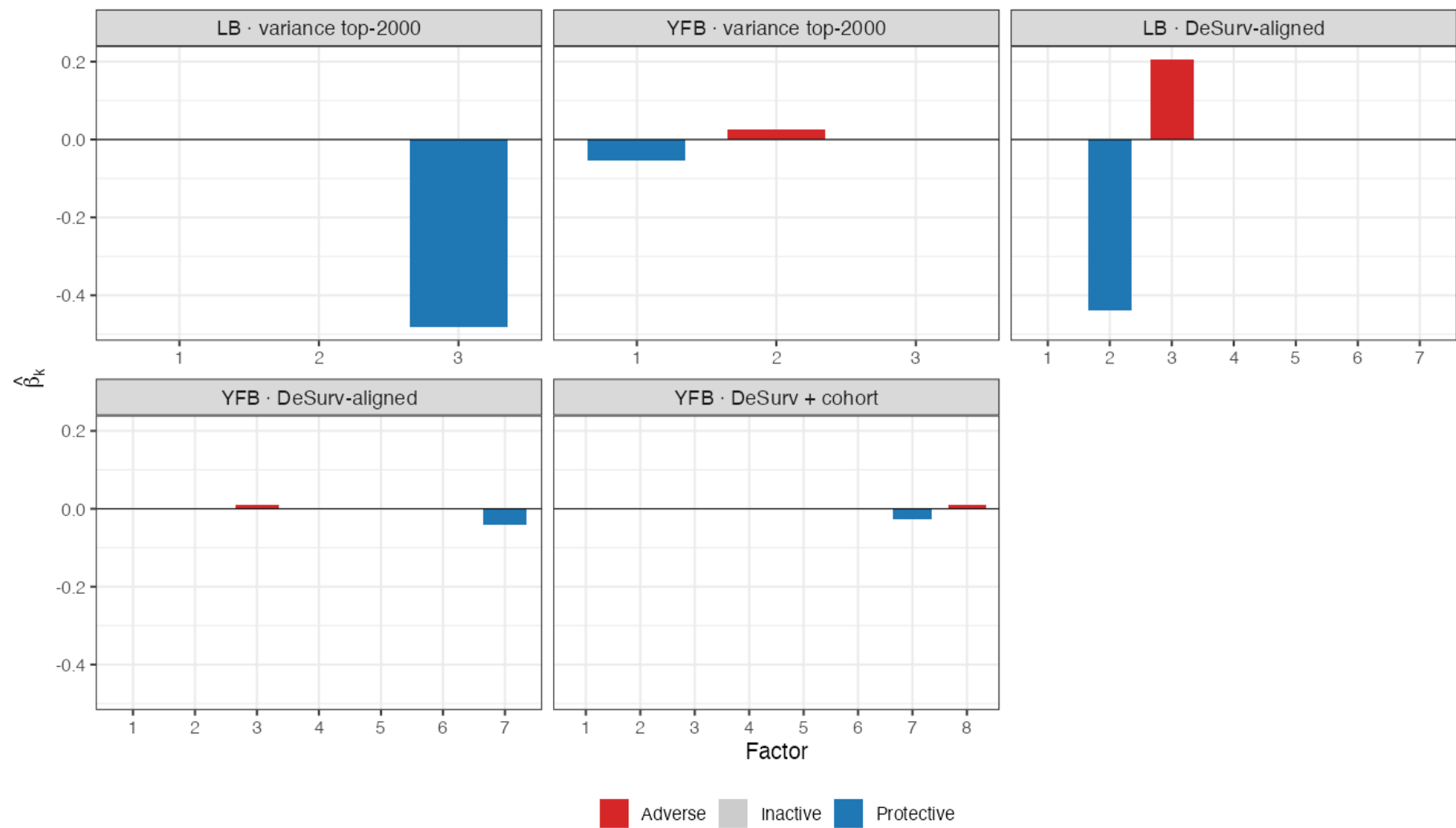


Figure 10: Posterior survival coefficients $\hat{\beta}$ per program

- RMSE stabilizes to a plateau; the dual ΔL & Δ $< 1e-3$ criterion governs stopping
- $\hat{\beta}$ barplots show the spike-and-slab at work: most programs shrunk to zero, the survival-active ones retained

Appendix — synthetic factor structure

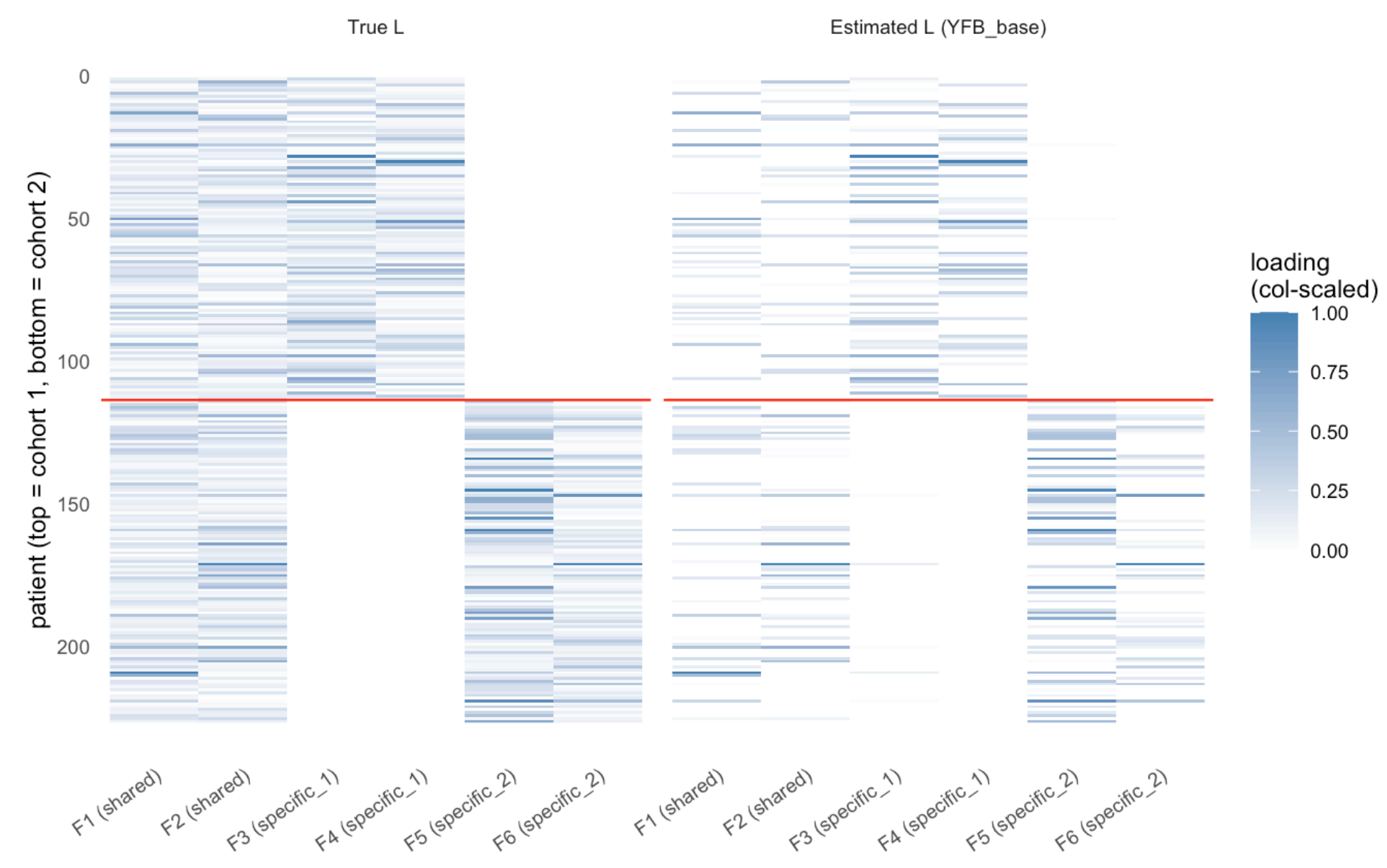


Figure 11: Estimated vs true loadings

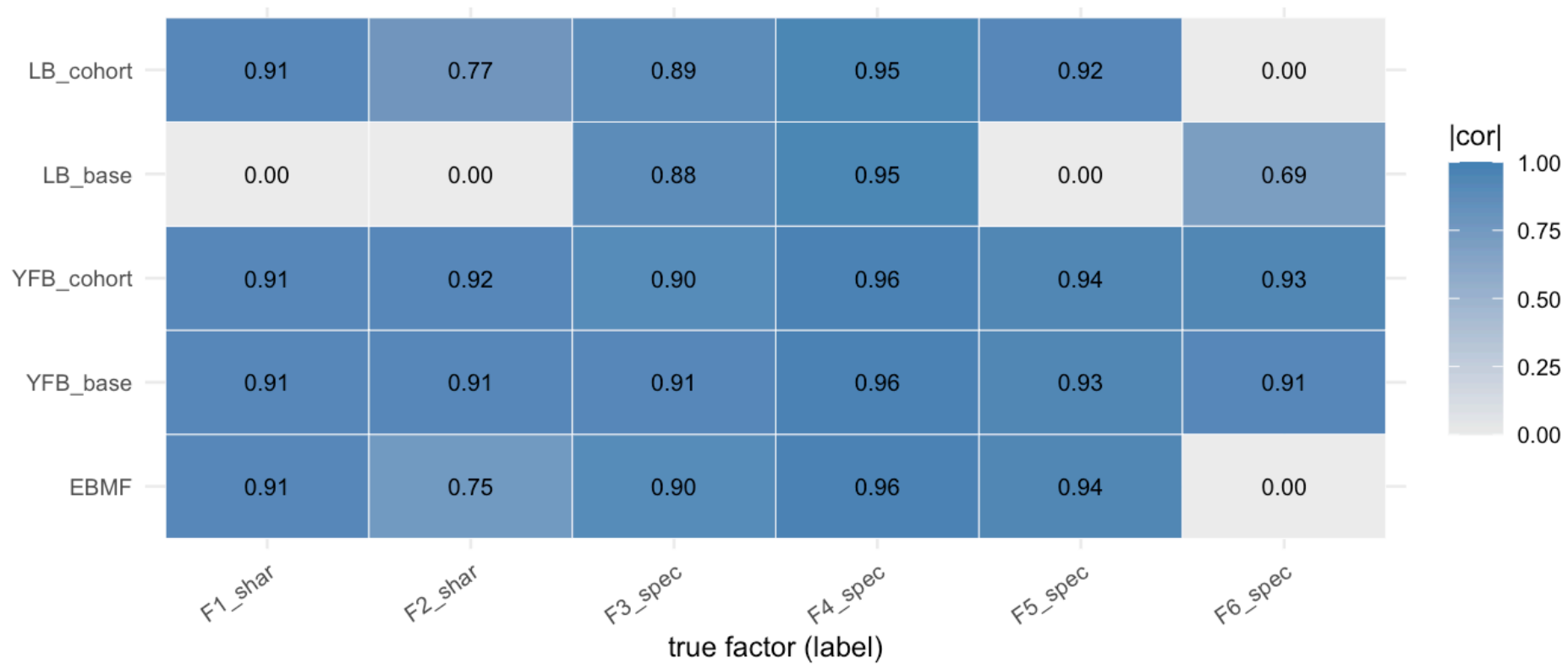


Figure 12: Factor heatmap (shared vs specific)

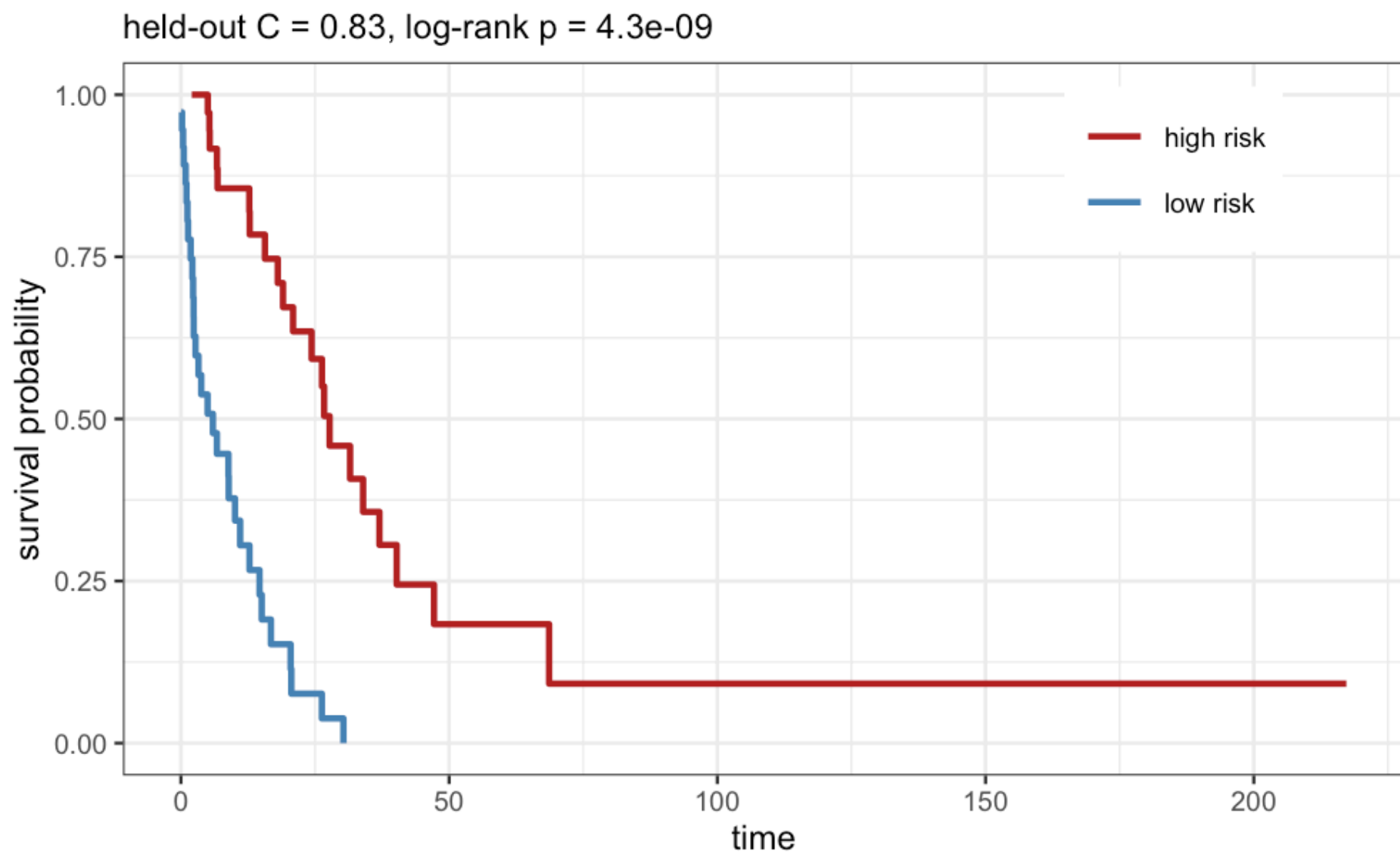


Figure 13: Synthetic survival stratification

- Ground-truth recovery: shared factors (survival-bearing) cleanly separated from study-specific factors
- The simulation establishes that the projection model recovers the *intended* latent structure, not merely a good survival metric